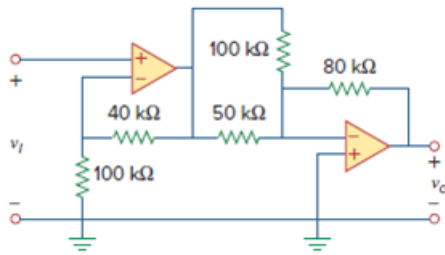


Problem

Calculate v_o/v_i in the op amp circuit of Fig. 5.87.

Figure. 5.87:



Step-by-step solution

Step 1 of 3

Refer to Figure 5.87 in the textbook and assign nodes, the modified circuit is shown in Figure 1.

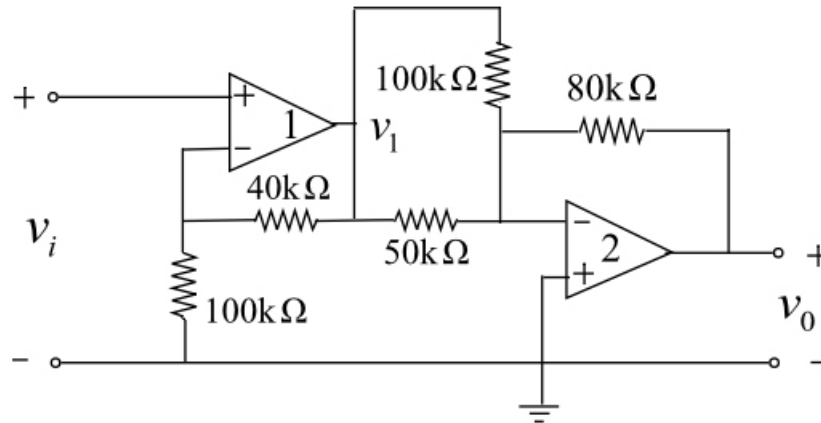


Figure - 1

Step 2 of 3

In Figure 1, the first op-amp is act as a non-inverting amplifier.

The output voltage for non-inverting amplifier is,

$$\begin{aligned} v_1 &= \left(1 + \frac{R_f}{R_i} \right) v_i \\ &= \left(1 + \frac{40 \times 10^3}{100 \times 10^3} \right) v_i \\ v_1 &= 1.4 v_i \end{aligned}$$

Step 3 of 3

In Figure 1, the second op-amp is act as an inverting amplifier.

The output voltage for inverting amplifier is,

$$\begin{aligned}v_1 &= -\left(\frac{R_f}{R_i}\right)v_i \\&= -\left(\frac{80 \times 10^3}{100 \times 10^3 \parallel 50 \times 10^3}\right)v_i \\&= -\left(\frac{80 \times 10^3}{\frac{(100 \times 10^3)(50 \times 10^3)}{100 \times 10^3 + 50 \times 10^3}}\right)v_i \\&= -\left(\frac{80 \times 10^3}{\frac{100 \times 10^3}{3}}\right)v_i \\&= -\left(\frac{240}{100}\right)v_i\end{aligned}$$

$$v_1 = -2.4v_i$$

Substitute $1.4v_i$ for v_1 .

$$v_o = -2.4(1.4v_i)$$

$$v_o = -3.36v_i$$

$$\frac{v_o}{v_i} = -3.36$$

Therefore, the voltage gain, $\frac{v_o}{v_i}$ is -3.36.

[Comments \(1\)](#)



Anonymous

This answer assumes the output of op amp 1 is the same as the input to op amp 2 which is wrong. The final answer is incorrect.
